

1.6 Solving Graphically

Formulate $\rightarrow$ graph constraints $\rightarrow$ feasible region $\rightarrow$ corner points $\rightarrow$ evaluate the objective.

Question 1 — Product Mix

A company produces two products, Product A and Product B. Each unit of Product A earns a profit of \$50, while each unit of Product B earns a profit of \$40.

Production is limited by the following two resource constraints:

$$A + 2B \leq 12$$

$$3A + B \leq 15$$

and production cannot be negative.

1. Let

2. Model

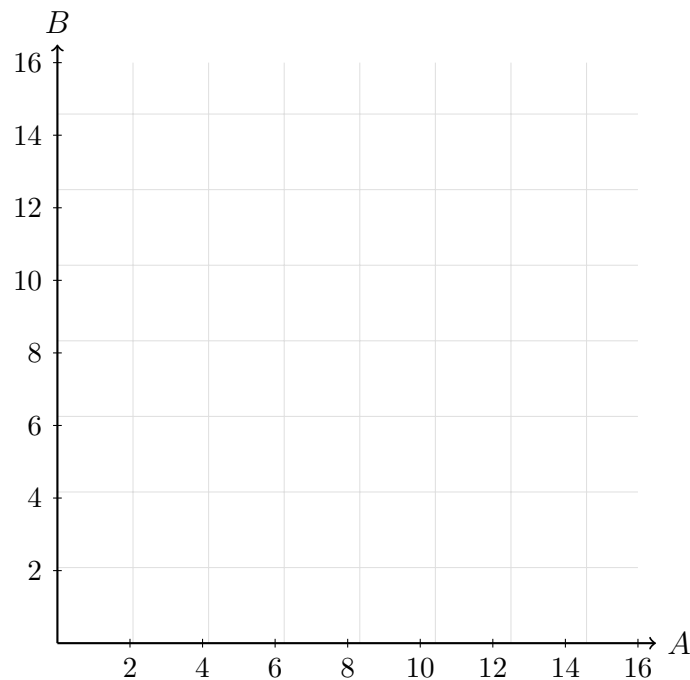
Decision variable	Money
A : Product A	\$50
B : Product B	\$40

$$\begin{aligned} \max Z = & \text{_____} A + \text{_____} B \\ \text{subject to (s.t.)} \end{aligned}$$

$$A + 2B \leq 12 \quad [1]$$

$$3A + B \leq 15 \quad [2]$$

$$A, B \geq 0 \quad [3]$$



Intercepts

[1] : (____, 0), (0, ____)

[2] : (____, 0), (0, ____)

Intersection of [1] and [2]

(A, B) = (____, ____)

Corner points

{(-, -), (-, -), (-, -), (-, -)}

Corner	$Z = 50A + 40B$
(____, ____)	
(____, ____)	
(____, ____)	
(____, ____)	

Optimal corner: (____, ____)

Question 2 — Multiple Optimal Solutions

A company produces two services, X and Y . Its objective is to maximize

$$Z = 2X + 4Y$$

subject to

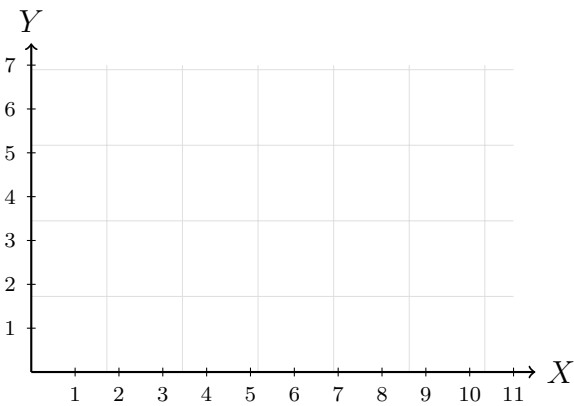
$$X + 2Y \leq 10, \quad X + Y \leq 7, \quad X, Y \geq 0.$$

1. Let

2. Model

Decision-variable	unit
X: Service X	2
Y: Service Y	4

$\max Z = 2X + 4Y$
subject to (s.t.)
 $X + 2Y \leq 10$ [1]
 $X + Y \leq 7$ [2]
 $X, Y \geq 0$ [3]



Intersection

Evaluate the objective

$$(X, Y) = (\text{____}, \text{____})$$

Corner points

$$\{(-, -), (-, -), (-, -), (-, -)\}$$

Two different corner points achieve the same best objective value. Check the slope of the edge joining them:

$$m = \frac{\text{rise}}{\text{run}} = \text{____} = \text{____}$$

Rewrite the objective as a line:

$$2X + 4Y = Z \quad \implies \quad Y = \text{_____}$$

$$\text{Objective slope} = \text{_____}$$

Lecture note: If the objective line is parallel to a binding edge of the feasible region, every point on that edge can have the same optimal objective value.

e.g.,

Binding edge [2]

Objective line [4]

$$X + 2Y = 10$$

$$2X + 4Y = Z$$

$$Y = -\frac{1}{2}X + 5$$

$$Y = -\frac{1}{2}X + \frac{Z}{4}$$

$$m_{[2]} = -\frac{1}{2}$$

$$m_{[4]} = -\frac{1}{2}$$

$$m_{[2]} = m_{[4]} \Rightarrow \text{parallel}$$

Objective parallel to a binding edge $\implies$ multiple optimal solutions along that edge.

1.4-1.8 (minus sensitivity analysis [1.7]) Solving Linear Programs with Excel

Let → Model → Excel → Solver

Question 1 — Media Selection

BP&J recommends that advertising be restricted to five media:

1. Daytime television (DTV): 1 minute
2. Evening television (ETV): 30 seconds
3. Daily newspaper (DN): 1 banner ad
4. Sunday newspaper (SN): 1 page colour
5. Radio (R): 30 seconds

The following data are given:

Decision variable	Objective	Other Stuff involved		
	Exposure (EQU)	Reach (customers)	Cost / ad (\$)	Availability (max/month)
DTV	65	1000	1500	15
ETV	90	2000	3000	10
Daily Newspaper	40	1500	400	25
Sunday Newspaper	60	2500	1000	4
Radio	20	300	100	30

Campaign requirements:

- Total budget cannot exceed \$30,000.
- At least 10 TV commercials must be broadcast.
- At least 50,000 customers must be reached.
- No more than \$18,000 may be spent on TV.

1. Let

Decision	Reach	Cost	Max	Exposure
: DTV	1000	1500	15	65
: ETV	2000	3000	10	90
: Daily	1500	400	25	40
: Sunday	2500	1000	4	60
: Radio	300	100	30	20

2. Model

$\max Z =$ _____

subject to (s.t.)

_____ [_____]

_____ [_____]

_____ [_____]

_____ [_____]

_____ [Bounds]

3. Excel

Decision cells → SUMPRODUCT → Solver → Solve
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Solver / spreadsheet

Question 2 — Airline Revenue Management

An airline is selling tickets for flights to a particular destination. The flight will depart in three weeks. The airline can use up to six planes, each costing \$50,000 to hire. Each plane has:

- 37 First-class seats,
- 38 Business-class seats,
- 60 Economy-class seats.

Once a customer purchases a ticket, there is no cancellation.

For administrative reasons, three price levels are feasible in each class, and one price level must be chosen for each class in each week.

The available prices are:

Week	Class	Option 1	Option 2	Option 3
1	First	1200	1000	950
1	Business	900	800	600
1	Economy	500	300	200
2	First	1400	1300	1150
2	Business	1100	900	750
2	Economy	700	400	350
3	First	1500	900	850
3	Business	820	800	500
3	Economy	480	470	450

Demand depends on the selected price:

Week	Class	Option 1	Option 2	Option 3
1	First	10	15	20
1	Business	20	25	35
1	Economy	45	55	60
2	First	20	25	35
2	Business	42	45	46
2	Economy	50	52	60
3	First	30	35	40
3	Business	40	50	55
3	Economy	50	60	80

Develop an optimization model that determines the price level for each class in each week, the number of seats to sell, and the number of planes to hire in order to maximize profit.

1. Let

2. Model

Decision variable	Meaning
x_{cwp}	Tickets sold for class c , week w , price option p
y_{cwp}	1 if price option p is selected; 0 otherwise
n	Number of planes hired

$\max Z =$ _____

subject to (s.t.)

_____ [Demand]

_____ [One price]

Indices:

c : class

w : week

p : price option

_____ [First capacity]

_____ [Business capacity]

_____ [Economy capacity]

_____ [Plane limit]

_____ [Domains]

3. Excel

Decision cells → SUMPRODUCT → Solver → Solve

Solver / spreadsheet

Question 3 — Campus Printing Company

Campus Printing Company produces two products for the university: course packs and event posters. Management is currently reviewing next month's production plan. Last semester, the company produced an average of 2,000 course packs and 1,500 event posters per month. During that period, the accounting department reported an average profit of \$2.00 per course pack and \$11.50 per event poster. The sales manager has suggested that the company should focus almost entirely on posters because they appear to be substantially more profitable.

The controller, however, points out that the reported product costs include several expenses that will be paid next month regardless of the number of course packs or posters produced.

Course packs sell for \$18 each, while event posters sell for \$25 each.

The company's accounting and production information is summarized below.

	Course Pack	Event Poster
Selling price	\$18.00	\$25.00
Reported total cost	\$16.00	\$13.50
Reported accounting profit	\$2.00	\$11.50
Paper and ink	\$6.00	\$8.00
Hourly production labour	\$3.00	\$4.00
Allocated rent and administration	\$4.00	\$1.50
Allocated equipment depreciation	\$3.00	—
Production labour time	6 min	12 min
High-resolution printer time	3 min	2 min
Last semester's average production	2,000	1,500
Maximum expected demand next month	2,500	3,000

The company has 400 production labour-hours available next month and 150 hours of high-resolution printer time available.

The company's monthly rent is \$18,000, administrative salaries total \$12,000 per month, and equipment depreciation is \$7,500 per month. These amounts will be incurred next month regardless of the production plan.

The high-resolution printer was purchased three years ago for \$84,000.

University enrollment has increased by approximately 8% compared with last year, and management expects campus event activity to remain strong.

Assume that all units produced can be sold up to the stated demand limits.

Determine the production plan that maximizes next month's operating contribution.

1. Let

Variable	Meaning
C	_____
P	_____

Before modeling:

What is each unit actually worth?

2. Model

What should one additional unit contribute to the objective?

$$\text{Course Pack: } 18 - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$\text{Event Poster: } 25 - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$\max Z = \underline{\hspace{1cm}}C + \underline{\hspace{1cm}}P$$

subject to (s.t.)

$$\underline{\hspace{3cm}} \quad [\text{Labour}]$$

$$\underline{\hspace{3cm}} \quad [\text{Printer}]$$

$$\underline{\hspace{3cm}} \quad [\text{Course-pack demand}]$$

$$\underline{\hspace{3cm}} \quad [\text{Poster demand}]$$

$$\underline{\hspace{3cm}} \quad [\text{Nonnegative}]$$

Question 4 — Merton Truck Company

Merton Truck Company manufactures two specialized trucks, Model 101 and Model 102, at a single manufacturing plant.

Management has recently become concerned about the company's financial performance. During the first six months of the year, Merton produced approximately 1,000 Model 101 trucks and 1,500 Model 102 trucks each month.

At the current production level, some departments are operating at full capacity while others still have unused production capacity.

The sales manager believes that Model 101 should perhaps be discontinued. Model 101 sells for \$39,000, but the company's accounting system reports a total product cost greater than its selling price. Model 102 sells for \$38,000 and appears substantially more profitable under the company's reported accounting figures.

The controller disagrees. He argues that part of the reported cost of each truck consists of fixed manufacturing overhead that has simply been allocated across the units produced. These fixed costs will continue to be incurred in the short run regardless of the particular product mix.

The following cost information is available:

	Model 101	Model 102
Selling price	\$39,000	\$38,000
Direct materials	\$24,000	\$20,000
Direct labour:		
Engine assembly	\$1,200	\$2,400
Metal stamping	\$800	\$600
Final assembly	\$2,000	\$1,500
Variable overhead:		
Engine assembly	\$2,100	\$4,000
Metal stamping	\$2,400	\$2,000
Final assembly	\$3,500	\$2,500
Reported total product cost	\$40,205	\$35,930
Current monthly production	1,000	1,500

Production takes place in four departments: engine assembly, metal stamping, Model 101 assembly, and Model 102 assembly.

Each department has a limited number of machine-hours available each month. The machine-hour requirements are shown below.

Department	Model 101	Model 102	Available / month
Engine assembly	1	2	4,000
Metal stamping	2	2	6,000
Model 101 assembly	2	—	5,000
Model 102 assembly	—	3	4,500

Merton can sell as many trucks as it is able to produce.

Management would like to determine whether its current product mix should be changed and, in particular, whether the apparently unprofitable Model 101 should actually be discontinued. Determine the monthly product mix that maximizes Merton's short-run operating contribution.

1. Let

Variable	Meaning
x_1	_____
x_2	_____

What does producing +1 truck actually change?

2. Model

What does one additional truck contribute?

Model 101: $39000 - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Model 102: $38000 - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

$$\begin{aligned} \max Z &= \underline{\hspace{1cm}}x_1 + \underline{\hspace{1cm}}x_2 \\ \text{subject to (s.t.)} \end{aligned}$$

$\underline{\hspace{10cm}}$ [Engine assembly]

$\underline{\hspace{10cm}}$ [Metal stamping]

$\underline{\hspace{10cm}}$ [Model 101 assembly]

$\underline{\hspace{10cm}}$ [Model 102 assembly]

$\underline{\hspace{10cm}}$ [Nonnegative]

1.8 Advanced Linear Programs

Same process:

Let $\rightarrow$ Model $\rightarrow$ Excel $\rightarrow$ Solver

Question 1 — Choosing MMA Courses

Suppose you are choosing courses for the remainder of the MMA program. Each course has been assigned a usefulness score and a difficulty score.

Variable	Course	Usefulness	Difficulty
<i>A</i>	MGSC 661 – Multivariate Statistical Analysis	9	5
<i>B</i>	INSY 669 – Text Analytics	7	3
<i>C</i>	INSY 670 – Social Media Analytics	6	2
<i>D</i>	MGSC 673 – Introduction to AI & Deep Learning	10	5
<i>E</i>	INSY 674 – Enterprise Data Science	8	4
<i>F</i>	INSY 684 – Enterprise ML in Production	9	4
<i>G</i>	ORGB 660 – Managing Data Analytics Teams	5	1
<i>H</i>	BUSA 697 – Careers	3	1

You may take at most five courses.

At least three courses must be selected.

At least half of the courses selected must be technical courses. For this problem, courses *A*, *B*, *D*, *E*, and *F* are classified as technical.

If course *D* is selected, course *A* must also be selected.

Exactly one of courses *B* and *C* may be selected.

The average difficulty of the selected courses must be at least 3.

Course *H* must be selected because apparently optimization cannot save you from Careers.

Determine the set of courses that maximizes total usefulness.

1. Let

$$A, B, C, D, E, F, G, H \in \{0, 1\}$$

where

$$x = \begin{cases} 1 & \text{if the course is taken} \\ 0 & \text{otherwise} \end{cases}$$

2. Model

$$\begin{aligned} \max Z = & \text{_____} \\ \text{subject to (s.t.)} & \end{aligned}$$

$$\text{_____} \quad [\text{At most five}]$$

$$\text{_____} \quad [\text{At least three}]$$

$$\text{_____} \quad [\text{At least half technical}]$$

$$\text{_____} \quad [\text{D requires A}]$$

$$\text{_____} \quad [\text{Exactly one of B or C}]$$

$$\text{_____} \quad [\text{Average difficulty}]$$

$$\text{_____} \quad [\text{H must be selected}]$$

$$\text{_____} \quad [\text{Binary}]$$

Binary Variables

For $x, y \in \{0, 1\}$:

Event	At least one	At most one	Exactly one	Both	x requires y
Model	$x + y \geq 1$	$x + y \leq 1$	$x + y = 1$	$x = y = 1$	$x \leq y$

1.7 Sensitivity Analysis

Question 1 — Reading a Sensitivity Report

Consider the linear program

$$\begin{aligned} \max Z = 50A + 40B \quad \text{s.t.} \quad & A + 2B \leq 12 \quad [1] \\ & 3A + B \leq 15 \quad [2] \\ & A + B \leq 10 \quad [3] \\ & A, B \geq 0 \end{aligned}$$

After solving the model using Simplex LP, Excel produces the following Sensitivity Report.

Variable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$B\$4	Decision 1	3.6	0	50	70	30
\$C\$4	Decision 2	4.2	0	40	60	23.3333

Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$H\$10	Constraint 1 LHS	12	14	12	5.5	7
\$H\$11	Constraint 2 LHS	15	12	15	11	9
\$H\$12	Constraint 3 LHS	7.8	0	10	1E+30	2.2

(a) Optimal solution

From the report,

$$A = 3.6, \quad B = 4.2$$

and

$$Z = 50(3.6) + 40(4.2) = 348.$$

(b) Binding constraints

Compare Final Value with RHS:

$$[1] : 12 \text{ vs. } 12$$

$$[2] : 15 \text{ vs. } 15$$

$$[3] : 7.8 \text{ vs. } 10$$

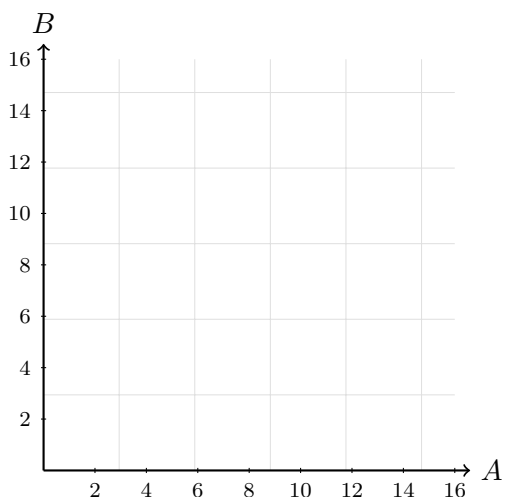
Therefore,

Binding: [1], [2] Nonbinding: [3]

(c) Slack

For constraint [3],

$$\text{Slack} = 10 - 7.8 = 2.2.$$



(d) Shadow Price Changing constraint RHS by 1, how much does it change z optimally? Look at constraint [1]:

$$A + 2B \leq 12.$$

Excel reports

$$\text{Shadow Price} = 14,$$

$$\text{Allowable Increase} = 5.5, \quad \text{Allowable Decrease} = 7.$$

If the RHS changes by +1,

$$12 \rightarrow 13,$$

the constraint line

but its slope

_____.

The feasible region therefore

_____.

The shadow price tells us

$$\Delta Z = 14(1) = \underline{\hspace{2cm}}.$$

So

$$Z_{\text{new}} = 348 + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}.$$

Constraint R.H. Side — Allowable Increase / Decrease

The current RHS is 12.

Excel gives

$$12 - 7 \leq \text{RHS} \leq 12 + 5.5.$$

Therefore,

$$\boxed{5 \leq \text{RHS} \leq 17.5}$$

Within this range, the shadow price remains

_____.

So within the range,

$$Z_{\text{new}} = 348 + 14(\Delta \text{RHS}).$$

Outside the range,

may change.

Graphically: move the constraint line parallel to itself.

(e) Objective Coefficient — Allowable Increase / Decrease

Now look at the objective:

$$Z = 50A + 40B.$$

Excel reports for A :

$$\text{Objective Coefficient} = 50,$$

$$\text{Allowable Increase} = 70, \quad \text{Allowable Decrease} = 30.$$

So

$$50 - 30 \leq c_A \leq 50 + 70,$$

giving

$$\boxed{20 \leq c_A \leq 120}$$

What changes on the graph?

The constraints:

The feasible region:

The objective line:

Rewrite the objective:

$$B = -\frac{c_A}{40}A + \frac{Z}{40}.$$

So changing c_A changes the

of the objective line.

As long as

$$20 \leq c_A \leq 120,$$

the optimal corner remains

Outside this range,

can become optimal.

(f) Reduced cost

From Excel,

$$\text{Reduced Cost of } A = 0, \quad \text{Reduced Cost of } B = 0.$$

Both variables are currently

in the optimal solution.

Reduced cost matters most when a variable has

$$\text{Final Value} = 0.$$

For a maximization problem, it tells us how much that variable's objective coefficient must

before it becomes worthwhile to enter the optimal solution.

Aside — What Is Excel Solver Actually Doing?

Return to the original two-constraint problem:

$$\max Z = 50A + 40B$$

$$A + 2B \leq 12, \quad 3A + B \leq 15, \quad A, B \geq 0.$$

We already solved it graphically.

What we did on the graph

Start somewhere we know works:

$$(A, B) = (0, 0), \quad Z = 0.$$

Move in the A direction.

$$A \leq 12 \quad [1]$$

$$A \leq 5 \quad [2]$$

So the first corner reached is

$$(0, 0) \longrightarrow (5, 0).$$

What Simplex records

Add slack:

$$A + 2B + s_1 = 12$$

$$3A + B + s_2 = 15.$$

Starting tableau:

	A	B	s_1	s_2	RHS
s_1	1	2	1	0	12
s_2	3	1	0	1	15
Z	-50	-40	0	0	0

Ratio test:

$$\frac{12}{1} = 12, \quad \frac{15}{3} = 5.$$

Same result:

$$(0, 0) \longrightarrow (5, 0).$$

First corner

At $(5, 0)$,

$$Z = 250.$$

Constraint [1]:

$$5 < 12 \quad \Rightarrow \quad s_1 = 7.$$

Constraint [2]:

$$15 = 15 \quad \Rightarrow \quad s_2 = 0.$$

So [2] is

binding.

But we already know

$$250 < 348,$$

so this is not the best corner yet.

After one Simplex iteration

	A	B	s_1	s_2	RHS
s_1	0	$\frac{5}{3}$	1	$-\frac{1}{3}$	7
A	1	$\frac{1}{3}$	0	$\frac{1}{3}$	5
Z	0	$-\frac{70}{3}$	0	$\frac{50}{3}$	250

There is still an improving move, so the algorithm continues.

$$(5, 0) \longrightarrow (3.6, 4.2).$$

What we found graphically

$$(0,0) \longrightarrow (5,0) \longrightarrow (3.6,4.2)$$

and

$$0 \longrightarrow 250 \longrightarrow 348.$$

At the last corner, there is no better feasible corner.

Final Simplex tableau

	<i>A</i>	<i>B</i>	s_1	s_2	RHS
<i>B</i>	0	1	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{21}{5}$
<i>A</i>	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{18}{5}$
<i>Z</i>	0	0	14	12	348

$$A = 3.6, \quad B = 4.2, \quad Z = 348.$$